Validated Spectral Analysis of the Gauss Transfer Operator and the Gauss–Kuzmin–Wirsing Law

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Abstract

We develop a rigorous Hardy-space framework for the Gauss transfer operator

$$(Lf)(w) = \sum_{n=1}^{\infty} \frac{1}{(w+n)^2} f\left(\frac{1}{w+n}\right),$$

acting from $H^2(D_1)$ to $H^2(D_{3/2})$. We prove compactness, derive explicit norm and truncation estimates, and implement a validated numerical scheme to enclose multiple eigenvalues and eigenvectors of L. This allows a certified computation of the Gauss–Kuzmin–Wirsing constant and a complete spectral expansion for continued-fraction digit distributions, answering Gauss's 1812 question with rigorous error bounds.

1 Introduction and Main Results

The Gauss map $T(x) = \{1/x\}$ on $(0, 1]$ generates continued fractions. Gauss asked for the probability

$$F_n(x) = \mathbb{P}(a_n \leq x)$$

that the n-th digit a_n of the continued-fraction expansion is at most x, if x_0 is uniformly distributed in $[0, 1]$. This distribution is governed by iterates of the transfer operator

$$(Lf)(x) = \sum_{n\geq 1} \frac{1}{(x+n)^2} f\left(\frac{1}{x+n}\right).$$

We work in Hardy spaces $H^2(D_r)$, with $D_r = \{|w-1| < r\}$, derive explicit analytic bounds, and rigorously enclose the spectrum of L.

Main results.

- $L : H^2(D_1) \to H^2(D_{3/2})$ is compact with approximation numbers $a_m(L) \leq C(2/3)^m$.
- Certified enclosures of eigenvalues and eigenvectors, including rigorous bounds on the Gauss–Kuzmin–Wirsing constant.
- Spectral expansion

$$F_n(x) = \frac{\ln(1+x)}{\ln 2} + \sum_{j\geq 2} c_j \lambda_j^n H_j(x), \quad H_j(x) = \int_0^x \phi_j(t)\, dt,$$

  with error $O(\rho^n)$.

2 Analytic Framework

Definition 2.1 (Hardy spaces)**.** *For $r > 0$, $H^2(D_r)$ denotes the space of holomorphic functions on $D_r = \{|w-1| < r\}$ with norm*

$$\|f\|^2_{H^2(D_r)} = \sum_{k\geq 0} |c_k|^2 r^{2k}, \qquad f(w) = \sum_{k\geq 0} c_k (w-1)^k.$$

Lemma 2.2 (Geometry of the branch maps)**.** *Let $\tau_n(w) = 1/(w+n)$. Then for all $n \geq 1$, $\tau_n(D_{3/2}) \subset D_1$.*

Proof. For $w \in D_{3/2}$, $\Re w > -\frac{1}{2}$. Compute

$$|\tau_n(w) - 1| = \frac{|w+n-1|}{|w+n|}, \qquad |w+n|^2 - |w+n-1|^2 = 2\Re w + (2n-1).$$

Since $\Re w > -\frac{1}{2}$, we get $2\Re w + (2n-1) > 2n-2 \geq 0$, with equality only for $n = 1$ and $w = -\frac{1}{2}$ on $\partial D_{3/2}$. Hence $|w+n-1| < |w+n|$ and $|\tau_n(w) - 1| < 1$, proving the claim. □

Proposition 2.3. *For each $n \in \mathbb{N}$, the composition operator*

$$C_{\tau_n} : H^2(D_1) \to H^2(D_{3/2}), \qquad C_{\tau_n} f = f \circ \tau_n,$$

satisfies

$$\|C_{\tau_n}\|_{H^2(D_1)\to H^2(D_{3/2})} \;\leq\; \left(\frac{1+|\phi_n(0)|}{1-|\phi_n(0)|}\right)^{1/2} = \sqrt{2n+1},$$

where $\phi_n = T_1^{-1} \circ \tau_n \circ T_{3/2}$ with $T_r(\zeta) = 1 + r\zeta$.

Proof. Let $U_r : H^2(D_r) \to H^2(\mathbb{D})$ be the unitary operator $(U_r f)(\zeta) = f(T_r(\zeta))$. Then

$$U_{3/2}\, C_{\tau_n}\, U_1^{-1} \;=\; C_{\phi_n}, \qquad \phi_n = T_1^{-1} \circ \tau_n \circ T_{3/2},$$

so $\|C_{\tau_n}\| = \|C_{\phi_n}\|$. A direct computation gives

$$\phi_n(\zeta) = \frac{-n - \frac{3}{2}\zeta}{n + 1 + \frac{3}{2}\zeta}, \qquad \phi_n(0) = -\frac{n}{n+1}.$$

By [3, Thm. 3.6], for $H^2(\mathbb{D})$ we have

$$\|C_\phi\|_{H^2(\mathbb{D})} \leq \left(\tfrac{1+|\phi(0)|}{1-|\phi(0)|}\right)^{1/2}.$$

Applying this to $\phi = \phi_n$ yields

$$\|C_{\tau_n}\|_{H^2(D_1)\to H^2(D_{3/2})} = \|C_{\phi_n}\|_{H^2(\mathbb{D})} \leq \left(\frac{1+\frac{n}{n+1}}{1-\frac{n}{n+1}}\right)^{1/2} = \sqrt{2n+1}.$$

□

Lemma 2.4. *For $w \in D_{3/2}$,*

$$|(w+n)^{-2}| \leq (n - \tfrac{1}{2})^{-2}.$$

Proof. The minimum of $|w+n|$ on $\overline{D_{3/2}}$ is at $w = -\frac{1}{2}$: $|w+n| \geq n - \frac{1}{2}$. □

Theorem 2.5. *We have*

$$\|L\|_{H^2(D_1)\to H^2(D_{3/2})} \;\le\; \sum_{n\ge 1} \frac{\sqrt{2n+1}}{(n-\frac{1}{2})^2} \;\le\; \sqrt{6}\,2^{3/2}\big(1-2^{-3/2}\big)\,\zeta\Big(\tfrac{3}{2}\Big) =: C_2.$$

Proof. From Lemma 2.4 and Proposition 2.3,

$$\|Lf\|_{H^2(D_1)\to H^2(D_{3/2})} \le \sum_{n\ge 1} \frac{1}{(n-\frac{1}{2})^2}\,\|C_{\tau_n}\|\,\|f\| \le \sum_{n\ge 1} \frac{\sqrt{2n+1}}{(n-\frac{1}{2})^2}\,\|f\|.$$

Use the elementary comparison $\frac{2n+1}{n-\frac{1}{2}} \le 6$ to get $\sqrt{2n+1} \le \sqrt{6}\,(n-\frac{1}{2})^{1/2}$.

Now, we observe that

$$\sum_{n\ge 1}(n-\tfrac{1}{2})^{-s} = \sum_{m\ge 0}(m+\tfrac{1}{2})^{-s} = 2^s \sum_{m\ge 0}(2m+1)^{-s},$$

and that

$$\sum_{m=0}^{\infty}(2m+1)^{-s} = \zeta(s) - \sum_{m=1}^{\infty}(2m)^{-s} = \zeta(s) - 2^{-s}\zeta(s) = (1-2^{-s})\,\zeta(s), \qquad \Re s > 1.$$

Thus:

$$\sum_{n\ge 1}(n-\tfrac{1}{2})^{-s} = 2^s(1-2^{-s})\,\zeta(s) = (2^s-1)\,\zeta(s).$$

Hence

$$\sum_{n\ge 1}\frac{\sqrt{2n+1}}{(n-\frac{1}{2})^2} \le \sqrt{6}\sum_{n\ge 1}(n-\tfrac{1}{2})^{-3/2} = \sqrt{6}\big(2^{3/2}-1\big)\,\zeta(\tfrac{3}{2}).$$

□

Remark 2.6. *We observe that*

$$\frac{\sqrt{2n+1}}{(n-\frac{1}{2})^2} = \sqrt{\frac{2n+1}{n-\frac{1}{2}}}\,(n-\tfrac{1}{2})^{-3/2}, \qquad \sup_{n\ge N}\frac{2n+1}{n-\frac{1}{2}} = 2 + \frac{2}{N-\frac{1}{2}}.$$

Hence, for any $N \ge 1$,

$$\|L\|_{H^2(D_1)\to H^2(D_{3/2})} \le \sum_{n=1}^{N-1}\frac{\sqrt{2n+1}}{(n-\frac{1}{2})^2} \;+\; \sqrt{2+\frac{2}{N-\frac{1}{2}}}\;\sum_{n\ge N}(n-\tfrac{1}{2})^{-3/2}.$$

Using the Hurwitz zeta, $\sum_{n\ge N}(n-\frac{1}{2})^{-3/2} = \zeta\big(\frac{3}{2},\, N-\frac{1}{2}\big)$.

In table 1 are rigorously computed upper bounds for C_2 as a function of the splitting N, $N=1$ is the base case from Theorem 2.5, using Arb, with maximum enclosure radius bounded by 5×10^{-34}.

N	$C_2(N)$
1	11.7000807086
2	10.4849507042
3	10.2709840576
4	10.1907129342
5	10.1506945125
10	10.0893118365
100	10.0590444185
1000	10.0581271764
10000	10.0580982915

Table 1: Refined upper bounds for C_2 as a function of the splitting N.

3 Truncation bounds and approximation numbers

Compare the analogous estimate in the H^∞–ellipse setting: [2, Thm. 3.3(ii)]. See also the published variant using the same factorization idea in [1].

Lemma 3.1. *Let $0 < r < R$ and let $f(w) = \sum_{n\geq 0} a_n(w-1)^n$ be holomorphic on $D_R = \{|w-1| < R\}$ with*

$$\|f\|^2_{H^2(D_R)} = \sum_{n\geq 0} |a_n|^2 R^{2n} < \infty.$$

Let $\Pi_k f := \sum_{n=0}^{k} a_n(w-1)^n$ be the degree-k truncation, and let $J : H^2(D_R) \to H^2(D_r)$ be the restriction map (inclusion at the level of Banach spaces). Then

$$\| Jf - \Pi_k f \|_{H^2(D_r)} \;\leq\; \left(\frac{r}{R}\right)^{k+1} \|f\|_{H^2(D_R)}.$$

In particular, for $r = 1$ and $R = \frac{3}{2}$,

$$\| Jf - \Pi_k f \|_{H^2(D_1)} \;\leq\; \left(\frac{2}{3}\right)^{k+1} \|f\|_{H^2(D_{3/2})}.$$

Proof. First, $\|J\|_{H^2(D_R)\to H^2(D_r)} = 1$: indeed, $\sum |a_n|^2 r^{2n} \leq \sum |a_n|^2 R^{2n}$ and equality holds for constants. By the definition of the $H^2(D_r)$ norm,

$$\|Jf - \Pi_k f\|^2_{H^2(D_r)} = \sum_{n>k} |a_n|^2 r^{2n} = \sum_{n>k} |a_n|^2 R^{2n} \left(\frac{r}{R}\right)^{2n}.$$

Since $0 < \frac{r}{R} < 1$ and $n > k$, we have $\left(\frac{r}{R}\right)^{2n} \leq \left(\frac{r}{R}\right)^{2(k+1)}$. Therefore

$$\|Jf - \Pi_k f\|^2_{H^2(D_r)} \leq \left(\frac{r}{R}\right)^{2(k+1)} \sum_{n>k} |a_n|^2 R^{2n} \leq \left(\frac{r}{R}\right)^{2(k+1)} \sum_{n\geq 0} |a_n|^2 R^{2n} = \left(\frac{r}{R}\right)^{2(k+1)} \|f\|^2_{H^2(D_R)},$$

and taking square roots gives the claim. □

Theorem 3.2. *Let $T := R\,L\,S : H^2(D_{3/2}) \to H^2(D_1)$, where S and R are the restriction maps $H^2(D_{3/2}) \to H^2(D_1)$. Let Π_k be the degree-k truncation on $H^2(D_{3/2})$ and π_K the degree-K truncation on $H^2(D_{3/2})$ (acting on the* range *of L). Then, with $C_2 := \|L\|_{H^2(D_1)\to H^2(D_{3/2})}$ (e.g. $C_2 \leq \sqrt{6}\, 2^{3/2}(1 - 2^{-3/2})\zeta(\frac{3}{2})$),*

$$\left\| T - R\,\pi_K\,L\,\Pi_k\,S \right\|_{H^2(D_{3/2})\to H^2(D_1)} \;\leq\; C_2 \left[\left(\frac{2}{3}\right)^{K+1} + \left(\frac{2}{3}\right)^{k+1} \right].$$

Proof. Decompose

$$T - R\,\pi_K\,L\,\Pi_k\,S = R\,(I-\pi_K)\,L\,S \;+\; R\,\pi_K\,L\,(I-\Pi_k)\,S =: A_K + B_k.$$

Since $\|R\| = \|S\| = 1$ and $\|L\| = C_2$, it suffices to bound the truncation tails after restricting from radius $3/2$ to radius 1.

For A_K (target-side tail): by Lemma 3.1 with $(r, R) = (1, 3/2)$ applied to $g \in H^2(D_{3/2})$,

$$\|R\,(I-\pi_K)g\|_{H^2(D_1)} \;\le\; \left(\frac{2}{3}\right)^{K+1} \|g\|_{H^2(D_{3/2})}.$$

Hence

$$\|A_K\| = \sup_{\|f\|_{H^2(D_{3/2})}=1} \|R\,(I-\pi_K)\,L\,Sf\|_{H^2(D_1)} \;\le\; \left(\frac{2}{3}\right)^{K+1} \|L\| = \left(\frac{2}{3}\right)^{K+1} C_2.$$

For B_k (domain-side tail): apply Lemma 3.1 to $f \in H^2(D_{3/2})$ to get

$$\|Sf - S\Pi_k f\|_{H^2(D_1)} \;\le\; \left(\frac{2}{3}\right)^{k+1} \|f\|_{H^2(D_{3/2})}.$$

Therefore

$$\|B_k\| \le \|R\|\,\|\pi_K\|\,\|L\|\,\|S(I-\Pi_k)\| \le C_2\left(\frac{2}{3}\right)^{k+1}.$$

Combining the two bounds yields the claim. □

Corollary 3.3 (One–sided finite–rank approximations of L_1 and $L_{3/2}$)**.** *Let $L : H^2(D_1) \to H^2(D_{3/2})$ be bounded with $\|L\| = C_2$, and set $L_1 = S\,L : H^2(D_1) \to H^2(D_1)$ and $L_{3/2} = L\,R : H^2(D_{3/2}) \to H^2(D_{3/2})$.*

(i) Range truncation for L_1. *For $K \ge 0$ define the finite–rank operator*

$$(L_1)_K := \Pi_K\,L_1 = \Pi_K\,S\,L \quad (H^2(D_1) \to H^2(D_1)).$$

Then

$$\|\,L_1 - (L_1)_K\,\|_{H^2(D_1)\to H^2(D_1)} \;\le\; C_2\left(\frac{2}{3}\right)^{K+1}.$$

(ii) Domain truncation for $L_{3/2}$. *For $k \ge 0$ define the finite–rank operator*

$$(L_{3/2})_k := L\,\Pi_k\,R \quad (H^2(D_{3/2}) \to H^2(D_{3/2})).$$

Then

$$\|\,L_{3/2} - (L_{3/2})_k\,\|_{H^2(D_{3/2})\to H^2(D_{3/2})} \;\le\; C_2\left(\frac{2}{3}\right)^{k+1}.$$

Proof. (i) For $f \in H^2(D_1)$, $L_1 f = S(Lf)$ is the restriction to D_1 of a function in $H^2(D_{3/2})$ with norm $\le C_2\|f\|_{H^2(D_1)}$. Apply Lemma 3.1 with $(r, R) = (1, 3/2)$ to $g := Lf \in H^2(D_{3/2})$:

$$\|\,L_1 f - \Pi_K L_1 f\,\|_{H^2(D_1)} = \|\,Sg - \Pi_K Sg\,\|_{H^2(D_1)} \le \left(\tfrac{2}{3}\right)^{K+1} \|g\|_{H^2(D_{3/2})} \le C_2\left(\tfrac{2}{3}\right)^{K+1} \|f\|_{H^2(D_1)}.$$

Taking the supremum gives the claim.

(ii) For $F \in H^2(D_{3/2})$,

$$\|\,L_{3/2}F - (L_{3/2})_k F\,\|_{H^2(D_{3/2})} = \|\,L(RF - \Pi_k RF)\,\|_{H^2(D_{3/2})} \le C_2\,\|\,RF - \Pi_k RF\,\|_{H^2(D_1)}.$$

Apply Lemma 3.1 with $(r, R) = (1, 3/2)$ to F to obtain $\|RF - \Pi_k RF\|_{H^2(D_1)} \le (2/3)^{k+1}\|F\|_{H^2(D_{3/2})}$, which yields the bound. □

Proposition 3.4. *Let $L : H^2(D_1) \to H^2(D_{3/2})$ be bounded with $\|L\| = C_2$, and set $L_1 = S\,L : H^2(D_1) \to H^2(D_1)$ and $L_{3/2} = L\,R : H^2(D_{3/2}) \to H^2(D_{3/2})$. Then:*

1. *L_1 and $L_{3/2}$ are compact.*
2. *For all $m \geq 1$,*
$$a_m(L_1) \;\leq\; C_2\Big(\tfrac{2}{3}\Big)^{m-1}, \qquad a_m(L_{3/2}) \;\leq\; C_2\Big(\tfrac{2}{3}\Big)^{m-1}.$$

Proof. By Corollary 3.3, for every $K \geq 0$ the rank-$\leq K+1$ operator $(L_1)_K = \Pi_K SL$ satisfies $\|L_1 - (L_1)_K\| \leq C_2(2/3)^{K+1}$; hence $\|L_1 - (L_1)_K\| \to 0$, so L_1 is compact. The same argument with $(L_{3/2})_k = L\Pi_k R$ gives compactness of $L_{3/2}$.

For the approximation numbers, fix $m \geq 1$. If $m = 1$, then $a_1(L_1) \leq \|L_1\| \leq C_2$ and similarly for $L_{3/2}$, which matches the stated bound. If $m \geq 2$, choose $K = k = m-2$ so that $\operatorname{rank}((L_1)_K) \leq m-1$ and $\operatorname{rank}((L_{3/2})_k) \leq m-1$. Then

$$a_m(L_1) \;\leq\; \|L_1 - (L_1)_{m-2}\| \;\leq\; C_2\Big(\tfrac{2}{3}\Big)^{(m-2)+1} \;=\; C_2\Big(\tfrac{2}{3}\Big)^{m-1},$$

and the same estimate holds for $a_m(L_{3/2})$. □

Remark 3.5. *An alternative proof of the compactness of $J_{R\to r}$ is to use the normal–family (Montel) theorem: the unit ball of $H^2(D_R)$ is locally bounded, hence normal; any bounded sequence has a subsequence converging uniformly on $\overline{D_r}$, and the coefficient tail estimate of Lemma 3.1 upgrades this to convergence in the $H^2(D_r)$–norm, yielding compactness.*

Corollary 3.6 (Eigenvalue decay and trace class for L_1 and $L_{3/2}$)**.** *Let $L_1 = SL : H^2(D_1) \to H^2(D_1)$ and $L_{3/2} = LR : H^2(D_{3/2}) \to H^2(D_{3/2})$ be as in Proposition 3.4. Enumerate the eigenvalues $\{\lambda_m(\cdot)\}_{m\geq 1}$ in nonincreasing order of modulus, with multiplicity. Then, with $C_2 = \|L\|_{H^2(D_1)\to H^2(D_{3/2})}$,*

$$|\lambda_m(L_1)| \;\leq\; C_2\Big(\tfrac{2}{3}\Big)^{m-1}, \qquad |\lambda_m(L_{3/2})| \;\leq\; C_2\Big(\tfrac{2}{3}\Big)^{m-1} \qquad (m \geq 1).$$

In particular, both L_1 and $L_{3/2}$ are trace class and satisfy the trace–norm bounds

$$\|L_1\|_{S_1} \;\leq\; 3\,C_2, \qquad \|L_{3/2}\|_{S_1} \;\leq\; 3\,C_2,$$

hence $\displaystyle\sum_{m\geq 1} |\lambda_m(L_1)| < \infty$ and $\displaystyle\sum_{m\geq 1} |\lambda_m(L_{3/2})| < \infty$.

Proof. By Proposition 3.4, the approximation numbers satisfy $a_m(L_1)$, $a_m(L_{3/2}) \leq C_2(2/3)^{m-1}$ for $m \geq 1$. On a Hilbert space, $a_m(T) = s_m(T)$ equals the mth singular value, and Weyl's inequality gives $|\lambda_m(T)| \leq s_m(T)$ (see Simon [4, Thm. 1.33]). Thus $|\lambda_m(L_1)|, |\lambda_m(L_{3/2})| \leq C_2(2/3)^{m-1}$.

Moreover, $\sum_{m\geq 1} s_m(L_1) \leq C_2 \sum_{m\geq 1}(2/3)^{m-1} = 3C_2$, and similarly for $L_{3/2}$. Hence L_1 and $L_{3/2}$ are trace class with the stated trace–norm bounds (see Simon [4, Thm. 1.35]). Finally, since $|\lambda_m| \leq s_m$, the series of absolute eigenvalues is summable. □

Remark 3.7. *Let A_k denote the Galerkin compression of either endomorphism $L_1 := SL : H^2(D_1) \to H^2(D_1)$ or $L_{3/2} := LR : H^2(D_{3/2}) \to H^2(D_{3/2})$ onto $\mathcal{V}_N = \operatorname{span}\{(w-1)^n\}_{n=0}^{N-1}$, with orthogonal projection taken in the $H^2(D_r)$ inner product. Then*

$$\|A_k\|_{(r)} \;\leq\; \|L_r\| \;\leq\; C_2, \qquad r \in \{1, \tfrac{3}{2}\}.$$

Indeed, $\|L_1\| \leq \|S\|\,\|L\| = C_2$ and $\|L_{3/2}\| \leq \|L\|\,\|R\| = C_2$ (since $\|R\| = \|S\| = 1$), and compressions do not increase the operator norm.

4 Spectral stability and projections from the discretization

Recall the endomorphisms

$$L_1 := S\,L : H^2(D_1) \to H^2(D_1), \qquad L_{3/2} := L\,R : H^2(D_{3/2}) \to H^2(D_{3/2}),$$

and their one–sided finite–rank approximants from Corollary 3.3:

$$(L_1)_K = \Pi_K S L, \qquad (L_{3/2})_k = L \Pi_k R,$$

with

$$\|L_1 - (L_1)_K\| \le \varepsilon_K := C_2 \Big(\tfrac{2}{3}\Big)^{K+1}, \qquad \|L_{3/2} - (L_{3/2})_k\| \le \delta_k := C_2 \Big(\tfrac{2}{3}\Big)^{k+1}.$$

We now record a quantitative "resolvent bridge" and its consequences for enclosing eigenvalues and approximating spectral projections.

Proposition 4.1. *Let (λ, φ) be a (nonzero) eigenpair of L with $\varphi \in H^2(D_1) \cap H^2(D_{3/2})$ and $L\varphi = \lambda\varphi$. Then*

$$\boxed{\|\varphi\|_{H^2(D_{3/2})} \;\le\; \frac{C_2}{|\lambda|}\,\|\varphi\|_{H^2(D_1)}.}$$

In particular, under the normalization $\|\varphi\|_{H^2(D_1)} = 1$,

$$\|\varphi\|_{H^2(D_{3/2})} \;\le\; \frac{C_2}{|\lambda|}.$$

Proof. Since $L\varphi = \lambda\varphi$ and $L : H^2(D_1) \to H^2(D_{3/2})$,

$$\|\varphi\|_{H^2(D_{3/2})} = \frac{1}{|\lambda|}\,\|L\varphi\|_{H^2(D_{3/2})} \le \frac{C_2}{|\lambda|}\,\|\varphi\|_{H^2(D_1)}.$$

□

Lemma 4.2 (Resolvent bridge)**.** *Let A, B be bounded operators on a Banach space and fix $z \in \mathbb{C}$ with $z \in \rho(B)$ (the resolvent set). If*

$$\alpha(z) := \|(A - B)R_B(z)\| \;<\; 1,$$

then $z \in \rho(A)$ and

$$\|R_A(z)\| \;\le\; \frac{\|R_B(z)\|}{1 - \alpha(z)}, \qquad \|R_A(z) - R_B(z)\| \;\le\; \frac{\|R_B(z)\|^2\,\|A - B\|}{1 - \alpha(z)}. \tag{1}$$

Moreover, if Γ is a positively oriented Jordan curve with $\Gamma \subset \rho(B)$ and $\sup_{z\in\Gamma} \alpha(z) < 1$, then $\Gamma \subset \rho(A)$ and the Riesz projections

$$P_A(\Gamma) = \frac{1}{2\pi i}\int_\Gamma R_A(z)\,dz, \qquad P_B(\Gamma) = \frac{1}{2\pi i}\int_\Gamma R_B(z)\,dz$$

are both well defined and satisfy

$$\|P_A(\Gamma) - P_B(\Gamma)\| \;\le\; \frac{|\Gamma|}{2\pi}\,\sup_{z\in\Gamma} \frac{\|R_B(z)\|^2\,\|A - B\|}{1 - \alpha(z)}, \tag{2}$$

and A and B have the same total algebraic multiplicity of eigenvalues inside Γ.

Proof. Use the resolvent identity $R_A(z) = R_B(z)\big[I - (A - B)R_B(z)\big]^{-1}$ and Neumann series when $\|(A - B)R_B(z)\| < 1$, then integrate on Γ for the projection bounds. □

Applications to L_1 and $L_{3/2}$

We apply Lemma 4.2 with

$$(A, B) \in \big\{(L_1, (L_1)_K),\ (L_{3/2}, (L_{3/2})_k)\big\}.$$

All norms below are operator norms on the indicated Hardy space.

If for some $z \in \mathbb{C}$ we have

$$\varepsilon_K \left\| R_{(L_1)_K}(z) \right\| \ < \ 1 \quad \text{or} \quad \delta_k \left\| R_{(L_{3/2})_k}(z) \right\| \ < \ 1,$$

then $z \in \rho(L_1)$ or $z \in \rho(L_{3/2})$, respectively, and

$$\|R_{L_1}(z)\| \le \frac{\|R_{(L_1)_K}(z)\|}{1 - \varepsilon_K \|R_{(L_1)_K}(z)\|}, \qquad \|R_{L_{3/2}}(z)\| \le \frac{\|R_{(L_{3/2})_k}(z)\|}{1 - \delta_k \|R_{(L_{3/2})_k}(z)\|}.$$

Thus validated upper bounds on the discretized resolvent certify **spectral gaps** for L_1 or $L_{3/2}$.

Let Γ be a simple closed contour containing a cluster of eigenvalues of the discretization (but no other spectrum), and suppose

$$\sup_{z \in \Gamma} \ \varepsilon_K \left\| R_{(L_1)_K}(z) \right\| \ < \ 1 \quad \text{or} \quad \sup_{z \in \Gamma} \ \delta_k \left\| R_{(L_{3/2})_k}(z) \right\| \ < \ 1.$$

Then Γ encloses exactly the same total algebraic multiplicity of eigenvalues for L_1 (resp. $L_{3/2}$) as for the discretization. In particular, taking Γ to be a small circle around a simple eigenvalue of the discretization yields a validated enclosure for the corresponding eigenvalue of L_1 or $L_{3/2}$.

Under the same hypothesis as in (B), the spectral projector onto the eigenspace(s) inside Γ satisfies

$$\|P_{L_1}(\Gamma) - P_{(L_1)_K}(\Gamma)\| \ \le \ \frac{|\Gamma|}{2\pi} \sup_{z \in \Gamma} \frac{\|R_{(L_1)_K}(z)\|^2 \, \varepsilon_K}{1 - \varepsilon_K \|R_{(L_1)_K}(z)\|},$$

and the analogous bound holds with $L_{3/2}$ and $(L_{3/2})_k$, with δ_k in place of ε_K. Hence eigenvectors and invariant subspaces (right/left) can be approximated by integrating the discretized resolvent along Γ with an explicit a posteriori error bound.

Remarks. (i) Since $(L_1)_K$ and $(L_{3/2})_k$ have finite rank, their resolvent norms on a circle Γ can be computed (and rigorously enclosed) by any certified linear–algebra routine (e.g., via Schur form and interval/ball arithmetic). (ii) Because the one–sided errors ε_K, δ_k decay like $(2/3)^{\cdot}$, taking modest K or k often suffices to meet the small–gain condition $\sup_{z \in \Gamma} \alpha(z) < 1$. (iii) If Γ surrounds the origin, note that the finite–rank maps may have 0 in their spectrum; choose Γ avoiding 0 or use two–sided compressions if needed (Thm. 3.2).

The resolvent identity and Riesz projections are standard; see, e.g., Kato, *Perturbation Theory for Linear Operators*, Ch. IV.

5 Schur bridge on $H^2(D_r)$

Fix $r > 0$ and work in the N–dimensional monomial subspace $\mathcal{V}_N = \operatorname{span}\{(w-1)^n\}_{n=0}^{N-1}$. In this basis the $H^2(D_r)$ Gram is diagonal,

$$S_r = \operatorname{diag}(r^{2n})_{n=0}^{N-1}, \qquad C_r := S_r^{1/2} = \operatorname{diag}(r^n)_{n=0}^{N-1}.$$

For a matrix $M \in \mathbb{C}^{N \times N}$ representing a linear map on $\mathcal{V}_N$, the $H^2(D_r)$–operator norm is

$$\|M\|_{(r)} := \| \, C_r M C_r^{-1} \, \|_2, \tag{3}$$

and the S_r–adjoint is $M^\sharp := S_r^{-1} M^* S_r$.

6 Schur build in $H^2(D_r)$ and resolvent bounds

Fix $r > 0$ and the N-dimensional monomial subspace $\mathcal{V}_N = \mathrm{span}\{(w-1)^n\}_{n=0}^{N-1}$. In this basis the $H^2(D_r)$ Gram is diagonal

$$S_r = \mathrm{diag}(r^{2n})_{n=0}^{N-1}, \qquad C_r := S_r^{1/2} = \mathrm{diag}(r^n)_{n=0}^{N-1},$$

and the induced $H^2(D_r)$-operator norm is

$$\|M\|_{(r)} := \| \, C_r \, M \, C_r^{-1} \, \|_2.$$

For matrices X, the S_r-adjoint is $X^\sharp := S_r^{-1} X^* S_r$.

Central procedure: build an $H^2(D_r)$-Schur pair

Given a discretization $A_k \in \mathbb{C}^{N\times N}$ of an endomorphism of $H^2(D_r)$, construct the Schur pair and its defects as follows.

1. **Whiten.** Set $\widetilde{A} := C_r \, A_k \, C_r^{-1}$. Then $\|\cdot\|_{(r)}$ becomes the Euclidean norm on $\widetilde{A}$.
2. **Euclidean Schur.** Compute the (complex) Schur decomposition

   $$\widetilde{A}\, Q = Q\, T,$$

   with Q unitary and T upper (quasi-)triangular.
3. **Unwhiten the Schur vectors.** Define

   $$X := C_r^{-1} Q, \qquad X^\sharp := S_r^{-1} X^* S_r.$$

   Then $A_k X = XT$ up to roundoff, and $X^\sharp X \approx I$.
4. **Defects to be measured (cheaply).**

   $$\delta \;:=\; \|I - X^\sharp X\|_{(r)} = \|I - Q^* Q\|_2, \qquad R \;:=\; A_k X - XT, \quad r_{\mathrm{res}} \;:=\; \|R\|_{(r)} = \|\widetilde{A} Q - QT\|_2.$$

   Set $S_0 := XTX^\sharp$ and $E := A_k - S_0$.

All quantities above are obtained directly from (Q, T) and diagonal scalings by C_r. In exact arithmetic, Q is unitary so $\delta = 0$; numerically δ is at machine scale.

Theorem 6.1 (From the Schur build to $H^2(D_r)$ resolvent bounds)**.** *With the objects from the procedure above, let $X = UH$ be the S_r-polar factorization ($U^\sharp U = I$, $H = (X^\sharp X)^{1/2}$). Then:*

$$\begin{aligned}
&\|X\|_{(r)} \le \sqrt{1+\delta}, \\
&\|X^{-1}\|_{(r)} \le (1-\delta)^{-1/2}, \\
&\|H\|_{(r)} \|H^{-1}\|_{(r)} \le \frac{1+\delta}{1-\delta}. \\
&\|E\|_{(r)} \;\le\; r_{\mathrm{res}}\sqrt{1+\delta} \;+\; \|A_k\|_{(r)}\, \delta \;. \\
&\|(zI - S_0)^{-1}\|_{(r)} \;\le\; \frac{1+\delta}{1-\delta}\, \|(zI - T)^{-1}\|_2,
\end{aligned}$$

And if

$$\alpha_r(z) := \|(zI - S_0)^{-1}\|_{(r)}\, \|E\|_{(r)} < 1$$

we have

$$\|(zI - A_k)^{-1}\|_{(r)} \;\le\; \frac{\dfrac{1+\delta}{1-\delta}\,\|(zI-T)^{-1}\|_2}{1-\alpha_r(z)}$$

Expanded proof. Recall $\|M\|_{(r)} := \|C_r M C_r^{-1}\|_2$ with $C_r = \mathrm{diag}(r^n)$ and $X^\sharp = S_r^{-1} X^* S_r$.

(1) Orthogonality defect. If $\|I - X^\sharp X\|_{(r)} = \delta < 1$, then (Loewner order)

$$(1-\delta)I \preceq X^\sharp X \preceq (1+\delta)I,$$

so

$$\|X\|_{(r)}^2 = \|X^\sharp X\|_{(r)} \le 1+\delta, \qquad \|X^{-1}\|_{(r)}^{-2} = \|(X^\sharp X)^{-1}\|_{(r)}^{-1} \ge 1-\delta,$$

hence

$$\|X\|_{(r)} \le \sqrt{1+\delta}, \qquad \|X^{-1}\|_{(r)} \le (1-\delta)^{-1/2}.$$

With the S_r–polar $X = UH$ ($U^\sharp U = I$, $H = (X^\sharp X)^{1/2}$) we get

$$\|H\|_{(r)}\|H^{-1}\|_{(r)} \le \frac{1+\delta}{1-\delta}.$$

(2) Similarity defect (add & subtract $A_k X X^\sharp$). Define $S_0 := XTX^\sharp$ and $E := A_k - S_0$. Then

$$E = \underbrace{A_k X X^\sharp - XTX^\sharp}_{(A_k X - XT)X^\sharp} \;+\; \underbrace{A_k - A_k X X^\sharp}_{A_k(I - XX^\sharp)}.$$

Taking $\|\cdot\|_{(r)}$ gives

$$\|E\|_{(r)} \le \|A_k X - XT\|_{(r)}\, \|X^\sharp\|_{(r)} + \|A_k\|_{(r)}\, \|I - XX^\sharp\|_{(r)}.$$

Since $\|X^\sharp\|_{(r)} = \|X\|_{(r)} \le \sqrt{1+\delta}$ and (see below) $\|I - XX^\sharp\|_{(r)} = \|I - X^\sharp X\|_{(r)} = \delta$, we obtain

$$\boxed{\|E\|_{(r)} \;\le\; r_{\mathrm{res}}\sqrt{1+\delta} \;+\; \|A_k\|_{(r)}\,\delta,}$$

with $r_{\mathrm{res}} := \|A_k X - XT\|_{(r)}$.

Why $\|I - XX^\sharp\|_{(r)} = \|I - X^\sharp X\|_{(r)}$. In whitened coordinates $\widetilde{X} := C_r X C_r^{-1}$,

$$\|I - XX^\sharp\|_{(r)} = \|I - \widetilde{X}\widetilde{X}^*\|_2, \qquad \|I - X^\sharp X\|_{(r)} = \|I - \widetilde{X}^*\widetilde{X}\|_2.$$

The nonzero eigenvalues of $\widetilde{X}\widetilde{X}^*$ and $\widetilde{X}^*\widetilde{X}$ are the same ($= \sigma_j(\widetilde{X})^2$), so the eigenvalues of $I-$those are both $\{1 - \sigma_j^2\}$; hence the spectral norms coincide.

(3) Resolvent comparison. With $X = UH$ we have $S_0 = U(HTH)U^\sharp$, and S_r–unitary conjugation preserves $\|\cdot\|_{(r)}$, so

$$\|(zI - S_0)^{-1}\|_{(r)} = \|(zI - HTH)^{-1}\|_{(r)} \le \|H\|_{(r)}\|H^{-1}\|_{(r)}\, \|(zI-T)^{-1}\|_2 \le \frac{1+\delta}{1-\delta}\,\|(zI-T)^{-1}\|_2.$$

(4) Resolvent of A_k. Write

$$zI - A_k = \big(I - (zI - S_0)^{-1}E\big)\,(zI - S_0).$$

If $\alpha_r(z) := \|(zI - S_0)^{-1}\|_{(r)}\, \|E\|_{(r)} < 1$, then Neumann's lemma gives

$$\|(zI - A_k)^{-1}\|_{(r)} \le \frac{\|(zI - S_0)^{-1}\|_{(r)}}{1 - \alpha_r(z)} \le \frac{\dfrac{1+\delta}{1-\delta}\,\|(zI-T)^{-1}\|_2}{1-\alpha_r(z)}.$$

□

7 Newton–Kantorovich framework on a single Hardy space and biorthogonality

Fix $r \in \{1, \frac{3}{2}\}$ and let L_r denote one of the endomorphisms

$$L_1 := S\,L : H^2(D_1) \to H^2(D_1), \qquad L_{3/2} := L\,R : H^2(D_{3/2}) \to H^2(D_{3/2}),$$

so that $\|L_r\| \le C_2$ and any Galerkin compression A_k satisfies $\|A_k\|_{(r)} \le \|L_r\| \le C_2$ (Remark 7.2 below). We assume $\lambda \in \sigma(L_r) \setminus \{0\}$ is *simple*.

Nonlinear map and biorthogonal normalization

For a bounded $M : H^2(D_r) \to H^2(D_r)$ define

$$F_M(z, v) \;:=\; \big(Mv - zv,\ \ell(v) - 1\big), \qquad (z, v) \in \mathbb{C} \times H^2(D_r),$$

with a fixed linear functional $\ell \in (H^2(D_r))^*$ used to pin phase/scale. We take ℓ from the *discretization*: if $A_k v_A = \lambda_A v_A$ and $u_A^\sharp A_k = \lambda_A u_A^\sharp$ are right/left eigenvectors, normalize by $\langle u_A, v_A\rangle_r = 1$ and set $\ell(\cdot) = \langle u_A, \cdot\rangle_r$. Thus $F_{A_k}(\lambda_A, v_A) = 0$ and $\ell(v_A) = 1$.

Exact inverse of the linearization at a simple eigenpair

Let $P_A := v_A \otimes u_A$ and $Q_A := I - P_A$. Then for the derivative at (λ_A, v_A),

$$DF_{A_k}(\lambda_A, v_A)[\delta z, \delta v] = \big((A_k - \lambda_A)\delta v - \delta z\, v_A,\ \langle u_A, \delta v\rangle_r\big).$$

If λ_A is simple, the restriction $(A_k - \lambda_A)|_{Q_A} : Q_A H^2(D_r) \to Q_A H^2(D_r)$ is invertible and

$$\boxed{DF_{A_k}(\lambda_A, v_A)^{-1}(a, b) = \big(\ -\langle u_A, a\rangle_r,\ \ b\, v_A + \underbrace{(A_k - \lambda_A)|_{Q_A}^{-1} Q_A a}_{=:\, S_A a}\ \big).}$$

Hence, with the product norm $\|(a, b)\| := \|a\|_{(r)} + |b|$,

$$\|DF_{A_k}^{-1}\| \;\le\; \max\big\{\ \|u_A\|_{(r)} + \|S_A\|\ ,\ \|v_A\|_{(r)}\ \big\}.$$

If Γ is a circle centered at λ_A that contains no other spectrum of A_k, then the *reduced resolvent* obeys

$$S_A = \frac{1}{2\pi i}\int_\Gamma \frac{1}{z - \lambda_A}\,(zI - A_k)^{-1}\,dz, \qquad \Rightarrow \qquad \boxed{\|S_A\| \;\le\; \sup_{z\in\Gamma} \|(zI - A_k)^{-1}\|_{(r)}\ .}$$

Newton–Kantorovich transfer to L_r (same space)

Let (λ, v) be the (normalized) simple eigenpair of L_r with $\ell(v) = 1$ and $L_r v = \lambda v$. Consider the Newton maps $N_{A_k} := (\mathrm{id} - DF_{A_k}^{-1} F_{A_k})$ and N_{L_r}. If N_{A_k} is a contraction on a ball around (λ_A, v_A) with constant $\kappa < 1$, then (telescoping at the fixed points and using $F_{L_r}(\lambda, v) = 0$) we get

$$\big\|(\lambda_A, v_A) - (\lambda, v)\big\| \;\le\; \frac{1}{1 - \kappa}\ \|DF_{A_k}^{-1}\|\ \|(A_k - L_r)v\|_{(r)}.$$

Thus, using $\|(A_k - L_r)v\| \le \|A_k - L_r\|\,\|v\|$,

$$\boxed{\big\|(\lambda_A, v_A) - (\lambda, v)\big\| \;\le\; \frac{\|DF_{A_k}^{-1}\|}{1-\kappa}\,\|A_k - L_r\|_{(r)}\,\|v\|_{(r)}.}$$

Now,

In particular, if A_k is a Galerkin compression produced by range/domain truncation, then (Corollary 3.3)

$$\|A_k - L_1\|_{(1)} \;\le\; C_2\Big(\tfrac{2}{3}\Big)^{K+1}, \qquad \|A_k - L_{3/2}\|_{(3/2)} \;\le\; C_2\Big(\tfrac{2}{3}\Big)^{k+1},$$

and the right-hand side decays geometrically with K (or k).

Remark 7.1 (Using resolvent bounds for $\|DF_{A_k}^{-1}\|$)**.** *If Γ is a circle around λ_A with no other eigenvalues of A_k, then*

$$\|DF_{A_k}^{-1}\| \;\le\; \max\Big\{\; \|u_A\|_{(r)} + \sup_{z\in\Gamma}\|(zI - A_k)^{-1}\|_{(r)}\;,\; \|v_A\|_{(r)}\;\Big\}.$$

Both $\|v_A\|$ and $\|u_A\|$ are computable from the Schur build in $H^2(D_r)$, and the resolvent supremum on Γ admits certified bounds via Theorem 6.1.

Remark 7.2 (Norm control of the discretizations)**.** *If A_k is the $H^2(D_r)$-orthogonal compression of L_r onto $\mathcal{V}_N = \operatorname{span}\{(w-1)^n\}_{n=0}^{N-1}$, then*

$$\|A_k\|_{(r)} \;\le\; \|L_r\| \;\le\; C_2, \qquad r \in \{1, \tfrac{3}{2}\}.$$

Indeed, $\|L_1\| \le \|S\|\,\|L\| = C_2$ and $\|L_{3/2}\| \le \|L\|\,\|R\| = C_2$ with $\|R\| = \|S\| = 1$, and orthogonal compression does not increase the operator norm.

Replacing $\|v\|_{(r)}$ in the NK bound. In the single–space framework of Section 7 we derived

$$\big\|(\lambda_A, v_A) - (\lambda, v)\big\| \;\le\; \frac{\|DF_{A_k}^{-1}\|}{1-\kappa}\,\|A_k - L_r\|_{(r)}\,\|v\|_{(r)}. \tag{$*$}$$

There are two convenient ways to control $\|v\|_{(r)}$:

(A) Same–space normalization (recommended). Normalize the true eigenvector in the working space: $\|v\|_{(r)} = 1$. Then $(*)$ becomes

$$\big\|(\lambda_A, v_A) - (\lambda, v)\big\| \;\le\; \frac{\|DF_{A_k}^{-1}\|}{1-\kappa}\,\|A_k - L_r\|_{(r)}.$$

(B) Weak-to-strong conversion (when the target normalization is in $H^2(D_1)$). Suppose $r = \frac{3}{2}$ but you keep the weak normalization $\|v\|_{H^2(D_1)} = 1$. By Proposition 10.1,

$$\|v\|_{(3/2)} = \|v\|_{H^2(D_{3/2})} \;\le\; \frac{C_2}{|\lambda|}\,\|v\|_{H^2(D_1)} \;=\; \frac{C_2}{|\lambda|}.$$

Plugging this into $(*)$ with $r = \frac{3}{2}$ yields the one–norm estimate

$$\boxed{\big\|(\lambda_A, v_A) - (\lambda, v)\big\| \;\le\; \frac{C_2}{(1-\kappa)\,|\lambda|}\,\|DF_{A_k}^{-1}\|\,\|A_k - L_{3/2}\|_{(3/2)}.}$$

(The same argument shows $\|v\|_{(3/2)} \le (C_2/|\lambda|)\|v\|_{(1)}$ for any scaling.)

8 Polynomials of operators on $H^2(D_r)$

Fix $r \in \{1, \frac{3}{2}\}$ and write $L_r \in \mathcal{B}(H^2(D_r))$ for either endomorphism $L_1 = SL$ or $L_{3/2} = LR$ (so $\|L_r\| \le C_2$). Let A_k be a Galerkin compression on $\mathcal{V}_N = \operatorname{span}\{(w-1)^n\}_{n=0}^{N-1}$, and set the single–space discretization error

$$\varepsilon_r \;:=\; \|L_r - A_k\|_{(r)}.$$

8.1 Polynomial perturbation in one norm

Let $p(z) = \sum_{j=1}^{d} a_j z^j$ with $p(0) = 0$. A telescoping expansion gives

$$p(L_r) - p(A_k) = \sum_{j=1}^{d} a_j \sum_{\ell=0}^{j-1} A_k^{\ell}\,(L_r - A_k)\,L_r^{j-1-\ell},$$

hence

$$\boxed{\|p(L_r) - p(A_k)\|_{(r)} \;\le\; \varepsilon_r \underbrace{\sum_{j=1}^{d} |a_j| \sum_{\ell=0}^{j-1} \|A_k\|_{(r)}^{\ell}\,\|L_r\|_{(r)}^{j-1-\ell}}_{=:\ \mathfrak{C}_r(p;A_k,L_r)}.} \tag{4}$$

Crude but convenient: using $\|A_k\|_{(r)}, \|L_r\|_{(r)} \le C_2$,

$$\|p(L_r) - p(A_k)\|_{(r)} \;\le\; \varepsilon_r \sum_{j=1}^{d} |a_j|\, j\, C_2^{j-1}. \tag{5}$$

8.2 Resolvent transfer for polynomials (same space)

Lemma 8.1 (Single–space resolvent transfer)**.** *Fix a contour $\Gamma \subset \mathbb{C}$. Let*

$$M_r \;:=\; \sup_{z\in\Gamma} \|(zI - p(A_k))^{-1}\|_{(r)}, \qquad \varepsilon_{p,r} \;:=\; \|p(L_r) - p(A_k)\|_{(r)}.$$

If $\varepsilon_{p,r}\, M_r < 1$, then $\Gamma \subset \rho(p(L_r))$ and

$$\sup_{z\in\Gamma} \|(zI - p(L_r))^{-1}\|_{(r)} \;\le\; \frac{M_r}{1 - \varepsilon_{p,r} M_r}.$$

Sketch. Resolvent identity: $z - p(L_r) = \big[I - (p(L_r) - p(A_k))(z - p(A_k))^{-1}\big]\,(z - p(A_k))$. Apply Neumann's lemma when $\varepsilon_{p,r} M_r < 1$. □

8.3 Computing M_r from a Schur build of A_k

Use the $H^2(D_r)$ Schur procedure of Section 6: whiten A_k by C_r, compute the Schur pair $\widetilde{A}Q = QT$, set $X = C_r^{-1}Q$, $\delta = \|I - X^{\sharp}X\|_{(r)}$ and $r_{\mathrm{res}} = \|A_k X - XT\|_{(r)}$. Since p is a polynomial,

$$p(S_0) = X\,p(T)\,X^{\sharp}, \qquad S_0 := XTX^{\sharp}.$$

Define $E_p := p(A_k) - p(S_0)$. Arguing as in Theorem 6.1,

$$\|(zI - p(S_0))^{-1}\|_{(r)} \;\le\; \frac{1+\delta}{1-\delta}\,\|(zI - p(T))^{-1}\|_2,$$

and, by the same add–subtract trick used for E,

$$\|E_p\|_{(r)} \;\le\; \sqrt{1+\delta}\,\sum_{j=1}^{d}|a_j|\sum_{\ell=0}^{j-1}\|A_k\|_{(r)}^{\ell}\,r_{\mathrm{res}}\;\|T\|_2^{\,j-1-\ell} \;+\; \delta\,\|p(A_k)\|_{(r)}. \tag{6}$$

If $\alpha_p(z) := \|(zI - p(S_0))^{-1}\|_{(r)}\,\|E_p\|_{(r)} < 1$, then

$$\boxed{\|(zI - p(A_k))^{-1}\|_{(r)} \;\le\; \frac{\dfrac{1+\delta}{1-\delta}\,\|(zI - p(T))^{-1}\|_2}{1-\alpha_p(z)} \;=:\; M_r(z),}$$

and $M_r := \sup_{z\in\Gamma} M_r(z)$ is a certified bound for Lemma 8.1. In exact arithmetic Q is unitary in the whitened metric ($\delta = 0$, $r_{\mathrm{res}} = 0$) and $\|(zI - p(A_k))^{-1}\|_{(r)} = \|(zI - p(T))^{-1}\|_2$.

8.4 Deflation polynomials and isolation

Let $\{\lambda_i\}$ be the Schur–ordered eigenvalues of A_k (in $H^2(D_r)$). Suppose $\{\widehat{\lambda}_i\}_{i<j}$ are certified. Define the deflation polynomial

$$P_j(z) := \prod_{i=1}^{j-1}\Big(1 - \frac{z}{\widehat{\lambda}_i}\Big), \qquad \alpha_j := \frac{1}{P_j(\lambda_j)}, \qquad p(z) := \big(\alpha_j P_j(z)\big)^q,$$

with the minimal integer $q \ge 1$ that yields the desired order of vanishing at the already certified set while keeping $p(\lambda_j) \approx 1$. Then the spectrum of $p(A_k)$ has one point near 1 (image of λ_j) and the rest contracted toward 0. Any small circle $\Gamma = \{z : |z-1| = r\}$ that excludes the other images isolates the target.

8.5 Projectors commute with polynomials

If Γ encloses a single eigenvalue of A_k (or L_r), then the Riesz projector is invariant under polynomial change:

$$\Pi = \frac{1}{2\pi i}\int_{\Gamma}(z - A)^{-1}\,dz = \frac{1}{2\pi i}\int_{p(\Gamma)}(w - p(A))^{-1}\,dw,$$

provided p is injective on the enclosed component. Thus one may compute $\widehat{\Pi}_j$ from $p(A_k)$ and transfer it to $p(L_r)$ via Lemma 8.1; mapping back through the local inverse of p yields the certified eigenvalue and projector for L_r.

8.6 Practical pipeline on $H^2(D_r)$

1. **Schur build for A_k.** Whiten by C_r, compute $\widetilde{A}Q = QT$, set $X = C_r^{-1}Q$, measure δ, r_{res}.
2. **Choose p.** Deflate previously certified modes; take minimal q.
3. **Bound M_r.** Use (6) and the resolvent bound with $p(T)$ to certify $M_r = \sup_{z\in\Gamma}\|(zI - p(A_k))^{-1}\|_{(r)}$.
4. **Bound $\varepsilon_{p,r}$.** Use (4) (or the crude (5) if desired) with $\varepsilon_r = \|L_r - A_k\|_{(r)}$.

5. **Transfer.** Check $\varepsilon_{p,r} M_r < 1$. Then

$$\sup_{z\in\Gamma} \|(zI - p(L_r))^{-1}\|_{(r)} \;\le\; \frac{M_r}{1-\varepsilon_{p,r}M_r},$$

so $\Gamma \subset \rho(p(L_r))$ and the projector is well defined with matching multiplicity.

6. **Back–map.** Use the local inverse of p at the image of λ_j to enclose $\lambda_j \in \sigma(L_r)$ and recover the projector on $H^2(D_r)$.

Remark 8.2 (Constants you actually need)**.** *All inputs are single–space: C_2 (for crude bounds), the discretization error $\varepsilon_r = \|L_r - A_k\|_{(r)}$ (from your one–sided truncation bounds), the Schur defects δ, r_{res}, and the triangular resolvent $\|(zI - p(T))^{-1}\|_2$. No cross–norms or embeddings are required.*

9 Outlook

This framework couples Hardy-space analysis and validated numerics. The same operator L_s for complex s yields $\det(I - L_s)^{-1} = \zeta(2s)$, forming the analytic basis for a companion paper.

Remark 9.1 (Connection with the Markov and Lagrange spectra)**.** *The parameter s in the Gauss–Kuzmin–Wirsing operator*

$$(L_s f)(w) = \sum_{n\ge 1} \frac{1}{(w+n)^{2s}} f\left(\frac{1}{w+n}\right)$$

acts as a thermodynamic deformation of the Gauss map potential. At $s = 1$, the operator describes the statistical law of continued fractions and governs the convergence of the empirical distribution functions F_n. As s decreases towards $1/2$, the leading eigenvalue $\lambda_1(s) = e^{P(2s)}$ traces the topological pressure $P(2s)$, whose zero determines the Hausdorff dimension of sets of numbers with bounded continued fractions:

$$P(2D) = 0, \qquad D \approx 0.53128.$$

This critical value D marks the onset of the Markov–Lagrange spectrum. Hence the same analytic family L_s interpolates between the probabilistic regime ($s = 1$) and the geometric–Diophantine regime ($s = D$), linking the Gauss problem with the classical spectra of Diophantine approximation.

10 Appendix: Pseudospectral approach to the enclosure of eigenvalues strong-weak framework

Recall that $L : H^2(D_1) \to H^2(D_{3/2})$ is bounded with

$$\|L\|_{H^2(D_1)\to H^2(D_{3/2})} \;\le\; C_2.$$

(For instance, one can take C_2 from Theorem 2.5.)

Proposition 10.1 (Strong norm from weak norm for eigenvectors)**.** *Let (λ, φ) be a (nonzero) eigenpair of L with $\varphi \in H^2(D_1) \cap H^2(D_{3/2})$ and $L\varphi = \lambda\varphi$. Then*

$$\boxed{\|\varphi\|_{H^2(D_{3/2})} \;\le\; \frac{C_2}{|\lambda|}\,\|\varphi\|_{H^2(D_1)}.}$$

In particular, under the normalization $\|\varphi\|_{H^2(D_1)} = 1$,

$$\|\varphi\|_{H^2(D_{3/2})} \;\le\; \frac{C_2}{|\lambda|}.$$

Proof. Since $L\varphi = \lambda\varphi$ and $L : H^2(D_1) \to H^2(D_{3/2})$,

$$\|\varphi\|_{H^2(D_{3/2})} = \frac{1}{|\lambda|}\,\|L\varphi\|_{H^2(D_{3/2})} \le \frac{C_2}{|\lambda|}\,\|\varphi\|_{H^2(D_1)}.$$

□

We now combine Proposition 10.1 with a resolvent estimate to obtain an exclosure for $\sigma(L)$ from a finite-rank approximation L_k.

Proposition 10.2. *Let L_k be a finite-rank operator such that*

$$\|L - L_k\|_{H^2(D_{3/2})\to H^2(D_1)} \le \eta.$$

If $\lambda \in \sigma(L)$ and $\lambda \notin \sigma(L_k)$, then

$$\big\|(\lambda - L_k)^{-1}\big\|_{H^2(D_1)\to H^2(D_1)} \;\ge\; \frac{|\lambda|}{C_2\,\eta}.$$

Equivalently,

$$\sigma(L) \;\subset\; \left\{ z \in \mathbb{C} \;\middle|\; \big\|(z - L_k)^{-1}\big\|_{H^2(D_1)\to H^2(D_1)} \;\ge\; \frac{|z|}{C_2\,\eta} \right\}.$$

Consequently, any region where $\|(z - L_k)^{-1}\|_{H^2(D_1)} \le 1/\delta$ with $\delta > \frac{C_2\,\eta}{|z|}$ is free of eigenvalues of L.

Proof. Let (λ, φ) be an eigenpair with $\varphi \neq 0$ and $\varphi \in H^2(D_1) \cap H^2(D_{3/2})$. Then

$$(\lambda - L_k)\varphi = (L - L_k)\varphi,$$

so, assuming $\lambda \notin \sigma(L_k)$,

$$\varphi \;=\; (\lambda - L_k)^{-1}(L - L_k)\varphi.$$

Taking the $H^2(D_1)$-norm and using $\|L - L_k\|_{H^2(D_{3/2})\to H^2(D_1)} \le \eta$ gives

$$\|\varphi\|_{H^2(D_1)} \;\le\; \big\|(\lambda - L_k)^{-1}\big\|_{H^2(D_1)\to H^2(D_1)}\,\eta\,\|\varphi\|_{H^2(D_{3/2})}.$$

Apply Proposition 10.1 to bound $\|\varphi\|_{H^2(D_{3/2})} \le (C_2/|\lambda|)\,\|\varphi\|_{H^2(D_1)}$ and cancel $\|\varphi\|_{H^2(D_1)} \neq 0$ to obtain $1 \le \big\|(\lambda - L_k)^{-1}\big\|\;\eta\;(C_2/|\lambda|)$, i.e. $\big\|(\lambda - L_k)^{-1}\big\| \ge |\lambda|/(C_2\,\eta)$. The set inclusion and the exclosure statement follow by contraposition. □

Lemma 10.3 (Energy inequality for the resolvent equation)**.** *Let $(\mathcal{B}_s, \|\cdot\|_s) \hookrightarrow (\mathcal{B}_w, \|\cdot\|_w)$ be a Banach couple and suppose $P : \mathcal{B}_s \to \mathcal{B}_s$ satisfies the strong Lasota–Yorke estimate*

$$\|Pu\|_s \le a\,\|u\|_s + b\,\|u\|_w, \qquad a, b \ge 0.$$

Fix $z \in \rho(P)$ and let $u \in \mathcal{B}_s$ solve the resolvent equation

$$(zI - P)u = f, \qquad f \in \mathcal{B}_s.$$

Then the solution obeys the energy inequality

$$\boxed{(|z| - a)\,\|u\|_s \;\le\; b\,\|u\|_w \;+\; \|f\|_s.}$$

In particular, the strong norm of u is controlled by its weak norm and the data f.

Proof. From $(zI - P)u = f$ we have

$$|z|\,\|u\|_s \le \|Pu\|_s + \|f\|_s.$$

Applying the Lasota–Yorke bound gives

$$|z|\,\|u\|_s \le a\,\|u\|_s + b\,\|u\|_w + \|f\|_s,$$

and rearranging yields the claimed inequality. □

Proposition 10.4. *Let $(\mathcal{B}_s, \|\cdot\|_s) \hookrightarrow (\mathcal{B}_w, \|\cdot\|_w)$ with continuous embedding $i : \mathcal{B}_s \to \mathcal{B}_w$ of norm $E_{s\to w} = \|i\|_{s\to w}$. Assume $P, P_k : \mathcal{B}_s \to \mathcal{B}_s$ satisfy the Lasota–Yorke bounds*

$$\|Pu\|_s \le a\,\|u\|_s + b\,\|u\|_w, \qquad \|Pu\|_w \le M\,\|u\|_w,$$

and let $\delta := \|P - P_k\|_{s\to w}$. Suppose for $z \notin \sigma(P_k)$ we have a weak resolvent bound

$$M_k(z) := \|(zI - P_k)^{-1}\|_{w\to w} < \infty.$$

If

$$|z| > a + b\,M_k(z)\,\delta,$$

then $z \notin \sigma(P)$ and the resolvent $R(z) = (zI - P)^{-1}$ satisfies the quantitative estimate

$$\boxed{\|R(z)\|_{s\to s} \;\le\; \frac{b\,M_k(z)\,E_{s\to w} + 1}{|z| - a - b\,M_k(z)\,\delta}.}$$

Proof. Let $R_k(z) = (zI - P_k)^{-1}$ and fix $f \in \mathcal{B}_s$ with $\|f\|_s = 1$. Set $u = R(z)f$, so that $(zI - P)u = f$ and hence

$$u = R_k(z)f + R_k(z)(P - P_k)u.$$

Taking weak norms gives

$$\|u\|_w \le M_k(z)\,\|f\|_w + M_k(z)\,\|(P - P_k)u\|_w \le M_k(z)\,E_{s\to w} + M_k(z)\,\delta\,\|u\|_s.$$

Apply Lemma 10.3 to $(zI - P)u = f$:

$$(|z| - a)\,\|u\|_s \le b\,\|u\|_w + 1.$$

Combining the two inequalities yields

$$(|z| - a)\,\|u\|_s \le bM_k(z)\,E_{s\to w} + bM_k(z)\,\delta\,\|u\|_s + 1,$$

collecting terms,

$$\big(|z| - a - b\,M_k(z)\,\delta\big)\,\|u\|_s \le b\,M_k(z)\,E_{s\to w} + 1.$$

If $|z| > a + b\,M_k(z)\,\delta$, divide through to obtain the bound; taking the supremum over $\|f\|_s = 1$ gives the stated estimate. □

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